

JESÚS ALEJANDRO DOMÍNGUEZ REYES

Software Engineer · Backend (Java / Spring Boot, Python) · Data & AI

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SUMMARY

Software Engineer with 4 years of professional backend experience building and evolving production systems with Java/Spring Boot and Python. Experienced in REST/GraphQL APIs, secure authentication, external-service integration, large-scale data validation (PySpark, billions of records) and AI/ML model delivery. Strong foundation in OOP, design patterns, testing and performance optimization; comfortable owning features end to end and taking over critical components when needed.

PROFESSIONAL EXPERIENCE

AlgorithmIA — Backend Developer

Aug 2023 – Present

Python, Java/Spring Boot, PySpark, AWS Lambda, Cloudera, Great Expectations

- Designed and implemented a Python library for secure execution of salary-estimation models, separating license enforcement, input validation, model protection, usage auditing and business logic into independent components.
- Implemented signed-license validation using machine-specific data to decrypt protected model weights, and encrypted usage auditing with cryptographic hashes and chained daily records backed by blockchain integrity verification.
- Optimized the library to run in under 30 ms per execution, with roughly half of runtime in model inference and the rest in validation and security controls.
- Built the initial Great Expectations validation workflow for a standardized data-quality platform (Python, PySpark) covering enterprise tables with billions of records; extended it with QA requirements and incident-driven fixes.
- Identified Spark session configurations that improved query response times on large Hive and Impala datasets in Cloudera Machine Learning.
- Developed Python REST APIs for AI models and Spring Boot RESTful APIs for statistical information, improving response times; refactored AWS Lambda functions and a Streamlit results dashboard using the Builder Pattern.

Accionamos Tus Datos (ATD) — Backend Developer

Aug 2022 – Aug 2023

Java, Spring Boot, Grails/Groovy, MongoDB, MySQL, JWT, Firebase, AWS

- Built Spring Boot features to retrieve users' bank accounts and categorized transactions through an external financial-data provider, implementing Controller, Service, Repository and DTO layers with MongoDB persistence.
- Designed REST endpoints from scratch for transaction retrieval with multiple filtering criteria, using GraphQL where appropriate; applied dependency injection, AOP and centralized exception handling.
- Implemented JWT authentication and authorization (token generation, cryptographic signing, validation) and session management.
- Integrated external financial APIs and webhook-based asynchronous processing; investigated and resolved failures in async workflows.
- Enhanced an internal Java SDK (Retrofit) that wrapped external API calls into reusable domain methods; owned its maintenance and new requirements after the technical lead's departure while mentoring a junior teammate.
- Developed endpoints, services and persistence across three Grails/Groovy microservices for the first version of a banking transaction-categorization platform, tested with Spock.
- Built a Firebase push-notification service, web-scraping systems (including Amazon Rekognition), and MySQL-based statistical calculations for financial analysis; wrote unit tests with Mockito covering success, failure and edge cases.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, SQL, Groovy, C

Backend & Frameworks: Spring Boot, Flask, Grails, Node.js, Vue.js, REST APIs, GraphQL, Webhooks, JWT, Retrofit

Data & AI: PySpark, Great Expectations, Hive, Impala, Cloudera ML, Streamlit, Machine Learning, Neural Networks

Cloud & Databases: AWS Lambda, Amazon Rekognition, Microsoft Azure, Firebase, MySQL, MongoDB

Engineering Practices: OOP, Design Patterns, Software Architecture, Microservices, Unit Testing (Mockito, Spock), Refactoring, Git, Gradle, Scrum, Kanban

Spoken Languages: Spanish (Native), English (Intermediate)

EDUCATION & CERTIFICATIONS

B.E. Computational Systems Engineering — Escuela Superior de Cómputo (ESCOM), IPN

2017 – 2022

Microsoft Certified: Azure Fundamentals (AZ-900)

2021

PROJECTS & SOCIAL SERVICE

- Fingerprint Recognizer (ESCOM social service, 2021–2022): designed and trained neural networks in Python; evaluated accuracy through false-positive / false-negative analysis.
- Vital-Signs Monitoring App: hybrid application with Vue.js, Node.js and Arduino sensors.
- Conway's Game of Life: concurrent simulation using Java threads.